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Homework — Chapter 3: Interdependence and the Gains from Trade

1. Problems and Applications (Mankiw)

Problem 4 (Canada)

a) Opportunity cost of 1 car = 15 bushels of wheat; of 1 bushel = 1/15 car. They are reciprocals.

b) PPF is a straight line from 20 million cars to 300 million bushels. Consuming 10 million cars uses 5 million workers, so Canada can consume 150 million bushels.

c) Canada produces 20 million cars and trades 10 million for 200 million bushels, consuming 10 million cars and 200 million bushels (outside the PPF). Canada should accept, since 20 bushels per car is more than its opportunity cost of 15.

Problem 5 (England and Scotland)

a) Absolute advantage: England in scones (50 > 40), Scotland in sweaters (2 > 1). Opportunity cost of a sweater: England 50 scones, Scotland 20 scones. Comparative advantage: England in scones, Scotland in sweaters.

b) Scotland exports sweaters because its opportunity cost of sweaters is lower.

c) Yes, both still gain. A sweater would cost Scotland 40 scones, still below England's 50, so Scotland keeps its comparative advantage in sweaters. Trade at a price between 40 and 50 scones per sweater benefits both.

Problem 6 (Boston and Chicago)

a) Price of 1 white sock: 1 red sock in Boston, 2 red socks in Chicago.

b) Boston has the absolute advantage in both colors. Comparative advantage: Boston in white socks, Chicago in red socks (1/2 white vs. 1 white in Boston).

c) Boston exports white socks; Chicago exports red socks.

d) Between 1 and 2 red socks per white sock.

Problem 9 (True or false)

a) True. Gains from trade come from comparative advantage, which every country has.

b) False. No one can have a comparative advantage in every good, because a low opportunity cost in one good means a high one in the other.

c) False. A trade at a price between the two opportunity costs benefits both parties.

d) False. Only trades at prices between the parties' opportunity costs benefit both.

e) False. The country gains overall, but some individuals, such as workers in import-competing industries, may lose.

2. Quick Quizzes

1-b, 2-c, 3-d, 4-b, 5-a, 6-d, 7-b, 8-d

3. Handbook: "Interdependence and Trade"

1-a, 2-b, 3-c, 4-c, 5-b, 6-b, 7-b, 8-c, 9-a, 10-c, 11-d, 12-a, 13-d, 14-b, 15-b, 16-b, 17-a, 18-c, 19-a, 20-c